

Overview of Softmax function

Subject: Softmax function

$$\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

1.

$$\sigma(\mathbf{z} + \mathbf{c})_j = \frac{e^{z_j + c}}{\sum_{k=1}^K e^{z_k + c}} = \frac{e^{z_j} \cdot e^c}{\sum_{k=1}^K e^{z_k} \cdot e^c} = \frac{e^{z_j}}{\sum_{k=1}^K e^{z_k}} = \sigma(\mathbf{z})_j$$

2.

As $\beta \rightarrow \infty$, temperature goes to zero, $T = 1/\beta \rightarrow 0$, so eventually all differences become large (relative to a shrinking temperature), which gives another interpretation for the limit behavior.

3.

$$\sigma(0, 10) := \sigma(1(0, 10)) = (1/(1 + e^{10}), e^{10}/(1 + e^{10})) \approx (0.00005, 0.99995)$$

4.

In words, the softmax applies the standard exponential function to each element z_i of the input tuple $\mathbf{z}$ (consisting of K real numbers), and normalizes these values by dividing by the sum of all these exponentials. The normalization ensures that the sum of the components of the output vector $\sigma(\mathbf{z})$ is 1. The term "softmax" derives from the amplifying effects of the exponential on any maxima in the input tuple. For example, the standard softmax of $(1, 2, 8)$ is approximately $(0.001, 0.002, 0.997)$, which amounts to assigning almost all of the total unit weight in the result to the position of the tuple's maximal element (of 8). In general, instead of e a different base $b > 0$ can be used. As above, if $b > 1$ then larger input components will result in larger output probabilities, and increasing the value of b will create probability distributions that are more concentrated around the positions of the largest input values. Conversely, if $0 < b < 1$ then smaller input components will result in larger output probabilities, and decreasing the value of b will create probability distributions that are more concentrated around the positions of the smallest input values. Writing $b = e^\beta$ or $b = e^{-\beta}$ (for real β) yields the expressions:

5.

$$P_t(a) = \frac{\exp(-q_t(a)/\tau)}{\sum_{i=1}^n \exp(-q_t(i)/\tau)},$$

6.

Statistical mechanics In statistical mechanics, the softmax function is known as the Boltzmann distribution (or Gibbs distribution): the index set $1, \dots, k$ are the microstates of the system; the inputs z_i are the energies of that state; the denominator is known as the partition function, often denoted by Z ; and the factor β is called the coldness (or thermodynamic beta, or inverse temperature).

7.

However, if the difference is small relative to the temperature, the value is not close to the arg max. For example, a difference of 10 is small relative to a temperature of 100:

8.

History The softmax function was used in statistical mechanics as the Boltzmann distribution in the foundational paper Boltzmann (1868), formalized and popularized in the influential textbook Gibbs (1902). The use of the softmax in decision theory is credited to R. Duncan Luce, who used the axiom of independence of irrelevant alternatives in rational choice theory to deduce the softmax in Luce's choice axiom for relative preferences. In machine learning, the term "softmax" is credited to John S. Bridle in two 1989 conference papers, Bridle (1990a); and Bridle (1990b):

9.

Applications The softmax function is used in various multiclass classification methods, such as multinomial logistic regression (also known as softmax regression), multiclass linear discriminant analysis, naive Bayes classifiers, and artificial neural networks. Specifically, in multinomial logistic regression and linear discriminant analysis, the input to the function is the result of K distinct linear functions, and the predicted probability for the j th class given a sample tuple x and a weighting vector w is:

10.

Mathematical properties Geometrically the softmax function maps the Euclidean space $\mathbb{R}^K$ to the boundary of the standard $(K-1)$ -simplex, cutting the dimension by one (the range is a $(K-1)$ -dimensional simplex in K -dimensional space), due to the linear constraint that all output sum to 1 meaning it lies on a hyperplane. Along the main diagonal $(x, x, \dots, x)$, softmax is just the uniform distribution on outputs, $(1/n, \dots, 1/n)$: equal scores yield equal probabilities. More generally, softmax is invariant under translation by the same value in each coordinate: adding $\mathbf{c} = (c, \dots, c)$ to the inputs $\mathbf{z}$ yields $\sigma(\mathbf{z} + \mathbf{c}) = \sigma(\mathbf{z})$, because it multiplies each exponent by the same factor, e^c (because $e^{z_i + c} = e^{z_i} \cdot e^c$), so the ratios do not change:

11.

Gradients The softmax function is also the gradient of the LogSumExp function: $\frac{\partial}{\partial z_i} \text{LSE}(\mathbf{z}) = \frac{\exp(z_i)}{\sum_{j=1}^K \exp(z_j)} = \sigma(\mathbf{z})_i$, for $i = 1, \dots, K$, $\mathbf{z} = (z_1, \dots, z_K) \in \mathbb{R}^K$, where the LogSumExp function is defined as $\text{LSE}(\mathbf{z}) = \log \left(\sum_{j=1}^K \exp(z_j) \right)$. The gradient of softmax is thus $\frac{\partial}{\partial z_j} \sigma_i = \sigma_i (\delta_{ij} - \sigma_j)$.